



KILLARA HIGH SCHOOL

2010

**HIGHER SCHOOL CERTIFICATE
HALF YEARLY EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question

Total marks – 84

- Attempt questions 1 – 7
- All questions are of equal value

Total marks – 84

Attempt Questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 mark) Use a SEPARATE writing booklet

- a) Write the exact value of $\cos \frac{5\pi}{4}$ **1**
- b) Solve $|2x - 3| < 13$ **2**
- c) The interval PQ has endpoints P(2, 3) and Q(−3, 5). Find the coordinates of the point which divides the interval externally in the ratio 3 : 1. **2**
- d) The polynomial $P(x) = 2x^3 + ax^2 + x + 2$ has a factor $(2x + 1)$. Find the value of a . **2**
- e) Find the obtuse angle between the lines $3x - y + 5 = 0$ and $2x + 3y - 1 = 0$. Give your answer to the nearest degree. **3**
- f) Evaluate $\lim_{x \rightarrow 0} \frac{2 \sin 2x}{x}$. **2**

Question 2 (12 mark) Use a SEPARATE writing booklet

- a) The equation $x^3 - 2x^2 + 4x - 5 = 0$ has roots α, β, γ .
- (i) Write down the values of $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. 2
- (ii) Hence find the values of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$. 1
- b) Use the table of standard integrals to find the exact value of 2
- $$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx.$$
- c) The variable point $(2\cos\theta, 3\sin\theta)$ lies on a curve. Find the Cartesian equation of the curve. 2
- d) Using the substitution $u = x + 2$, find $\int \frac{x}{3} \sqrt{x+2} \, dx$. 2
- e) (i) Express $\sqrt{3} \cos x - \sin x$ in the form $R \cos (x + \alpha)$ where 2
- $$0 < \alpha < \frac{\pi}{2} \text{ and } R > 0.$$
- (ii) Hence, solve $\sqrt{3} \cos x - \sin x = 1$ for $0 \leq x \leq \frac{\pi}{2}$. 1

Question 3 (12 mark) Use a SEPARATE writing booklet

- a) Use method of mathematical induction to prove that 4

$$(1+1) + (2+3) + (3+5) + \dots + (n + (2n-1)) = \frac{1}{2}n(3n+1)$$

where n is a positive integer.

- b) The letters of the word *MOUSE* are to be rearranged.

- (i) How many arrangements are there which start with the letter M and end with the letter E ? 1

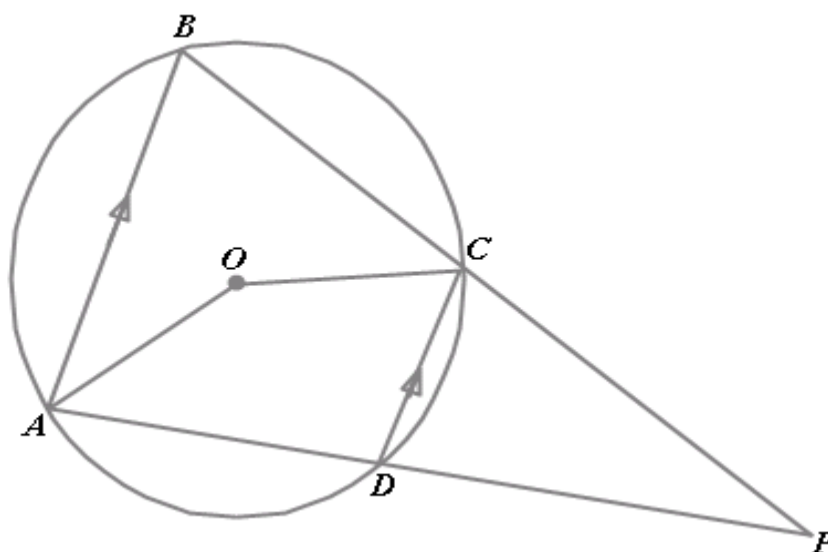
- (ii) How many arrangements are there in which the vowels are grouped together? 1
(A vowel is one of the letters A, E, I, O, U)

- (iii) How would the arrangements in part (i) and (ii) change if the given word had been *MOOSE* instead of *MOUSE*? 2

- c) Solve: $x + 2 \leq \frac{4}{x-1}$. 4

Question 4 (12 mark) Use a SEPARATE writing booklet

- a) (i) Show that the equation of the normal at $P(2at, at^2)$ on the parabola $x^2 = 4ay$ is $x + ty = at^3 + 2at$. 2
- (ii) The normal intersects with the y -axis at the point Q . Find the coordinates of Q and hence the coordinates of R , the midpoint of PQ . 2
- (iii) Hence, find the Cartesian equation of the locus of R . 1
- b) In the diagram below, O is the centre of the circle and $AB \parallel CD$. AD meets BC at P .



- Prove: (i) $CP = DP$. 3
- (ii) $\triangle ABP$ is isosceles. 2
- (iii) $OAPC$ is a cyclic quadrilateral. 2

Question 5 (12 mark) Use a SEPARATE writing booklet

a) Differentiate $\frac{\tan x}{x}$. **2**

b) Differentiate: $x \cos x$ **2**

c) Find $\int (\sin 3x + \cos 2x) dx$ **2**

d) Evaluate $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x dx$ **2**

e) Evaluate $\int_{-2\pi}^{2\pi} \cos^2 x \sin^3 x dx$ **1**

f) At any point $y = f(x)$, the gradient function is given by $\frac{dy}{dx} = 2 \cos^2 x - 1$. **3**
 If $y = \pi$ when $x = \pi$, find the value of y when $x = 2\pi$.

Question 6 (12 mark) Use a SEPARATE writing booklet

a) (i) Draw a neat sketch of $y = 3 \sin 2x$ for $0 \leq x \leq \pi$. **2**

(ii) The area between $y = 3 \sin 2x$ and, $x = 0$ and $x = \pi$ and the x - axis is rotated about the x - axis. Find the volume of the solid generated.
(Leave your answer in exact form.) **3**

b) The curve $x = \frac{4}{\sqrt[3]{y}}$ from $y = 1$ to $y = 8$ is revolved about the y -axis. **3**
Find the volume of the solid of revolution formed.

c) (i) Show that $\frac{d}{dx}(\tan^3 x) = 3 \sec^4 x - 3 \sec^2 x$. **2**

(ii) Hence, evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x \, dx$. **2**

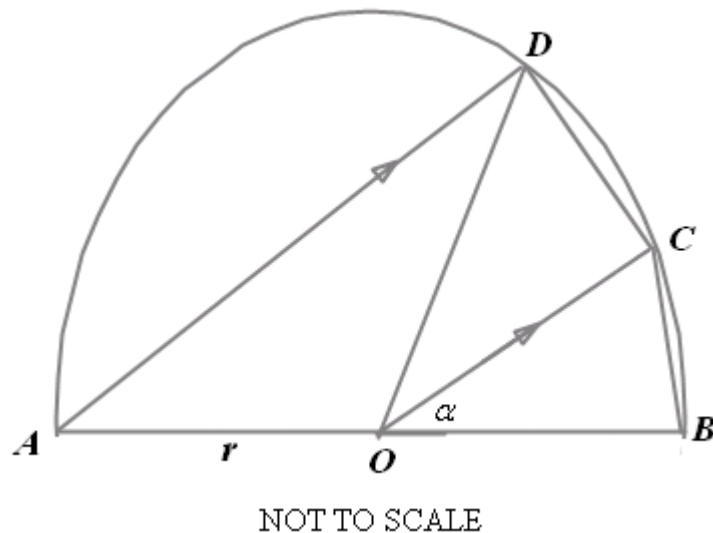
a) $y = \frac{(2x+1)^2}{4x(1-x)}$

(i) Find the equations of the horizontal and vertical asymptotes of this curve 3

(ii) Given that the curve has a relative maximum at $\left(-\frac{1}{2}, 0\right)$ and a relative minimum at $\left(\frac{1}{4}, 3\right)$, sketch the curve showing the asymptotes and turning points. 3

b) $ABCD$ is a quadrilateral inscribed in a semi circle with centre O and radius r .

$\angle COB = \alpha$ and AD is parallel to OC .



(i) Show that $\angle COD = \alpha$. Give reasons. 1

(ii) Show that the area of the quadrilateral $ABCD$ can be expressed as 2

$$A = r^2 \left(\frac{1}{2} \sin 2\alpha + \sin \alpha \right)$$

(iii) Find the value of α which will produce the quadrilateral of maximum area. 3

Question 1

(a)

Criteria	Marks
Gives the correct exact value of the trig. expression	1

$$\begin{aligned}\cos \frac{5\pi}{4} &= \cos \pi + \frac{\pi}{4} \\ &= -\cos \frac{\pi}{4} = -\frac{1}{\sqrt{2}} \quad (1 \text{ mark})\end{aligned}$$

(b)

Criteria	Marks
One mark for each correct solution	2

$$|2x - 3| < 13$$

$$-13 < 2x - 3 < 13$$

$$-10 < 2x < 16$$

$$-5 < x < 8$$

(c)

Criteria	Marks
Substitutes correctly into the formula	1
Correctly evaluates the coordinates	1

P(2, 3) and Q(-3, 5).
3 : 1

$$\text{The point is } \left(\frac{3 \times -3 - 1 \times 2}{3 - 1}, \frac{3 \times 5 - 1 \times 3}{3 - 1} \right) = \left(\frac{-11}{2}, 6 \right)$$

(d)

Criteria	Marks
• Substitutes $x = -\frac{1}{2}$ into $P(x) = 0$	1
• Correctly evaluates the value of a	1

$$P(x) = 2x^3 + ax^2 + x + 2$$

$$P\left(-\frac{1}{2}\right) = 0 \quad (1 \text{ mark})$$

$$2\left(-\frac{1}{2}\right)^3 + a\left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right) + 2 = 0$$

$$-\frac{1}{4} + \frac{1}{4}a - \frac{1}{2} + 2 = 0 \quad \therefore a = -5 \quad (1 \text{ mark})$$

(e)

Criteria	Marks
• Evaluates the correct values of m from each expression	2
• Correctly uses the formula and rounds off to the nearest degree	1

$$3x - y + 5 = 0 \Rightarrow y = 3x + 5 \quad \therefore m_1 = 3 \quad (1 \text{ mark})$$

$$2x + 3y - 1 = 0 \Rightarrow y = -\frac{2}{3}x + \frac{1}{3} \quad \therefore m_2 = -\frac{2}{3} \quad (1 \text{ mark})$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{3 - \left(-\frac{2}{3}\right)}{1 + 3 \times \left(-\frac{2}{3}\right)} \right| = \frac{11}{3}$$

$$\therefore \theta = 74^\circ 45'$$

$$\therefore \text{the obtuse angle is } 180 - 74^\circ 45' \approx 105^\circ \quad (1 \text{ mark})$$

(f)

Criteria	Marks
• Gives the correct solution	2
• Some attempt is made to manipulate the expression and identifies $\lim_{x \rightarrow 0} \frac{\sin 2x}{2x} = 1$	1
• Bald answer	0
• $2 \times 2 \lim_{x \rightarrow 0} \frac{\sin x}{x}$ (No fudging allowed)	0

$$\lim_{x \rightarrow 0} \frac{2 \sin 2x}{x} = \lim_{x \rightarrow 0} 2 \times 2 \frac{\sin 2x}{2x} = 2 \times 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} = 2 \times 2 \times 1 = 4$$

Question 2

Outcomes assessed: E2, E4

Criteria	Marks
- Both answers correct - One of them correct	2 1

- a) (i) $x^3 - 2x^2 + 4x - 5 = 0$
 $\alpha\beta + \beta\gamma + \gamma\alpha = 4$ (1 mark)
and $\alpha\beta\gamma = -(-5) = 5$ (1 mark)

Criteria	Marks
Uses the answers from part (i) to correctly evaluate the solution	1

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma} = \frac{4}{5}$

b)

Criteria	Marks
- Correctly identifies the integral of $\frac{1}{\sqrt{1-x^2}}$ as $\sin^{-1}x$ - Correctly evaluates the solution	1 1

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx = [\sin^{-1}(x)]_0^1 \quad (1 \text{ mark})$$

$$= \frac{\pi}{2} - 0 = \frac{\pi}{2} \quad (1 \text{ mark})$$

c)

Criteria	Marks
Gives correct expressions in terms of x and y for $\sin\theta$ and $\cos\theta$	1
Correctly eliminates $\sin\theta$ and $\cos\theta$	1

$$X = 2\cos\theta \rightarrow \cos\theta = \frac{x}{2}$$

$$Y = 3\sin\theta \rightarrow \sin\theta = \frac{y}{3}$$

$$\cos^2\theta + \sin^2\theta = 1$$

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

d)

Criteria	Marks
Uses the substitution to convert the integrand in terms of u.	1
Correct integration (ignore constant of integration)	1

$$u = x + 2 \rightarrow x = u - 2$$

$$dx = du$$

$$\int \frac{x}{3} \sqrt{x+2} dx = \int \frac{u-2}{3} \sqrt{u} du \quad (1 \text{ mark})$$

$$= \frac{1}{3} \int u^{\frac{3}{2}} - 2u^{\frac{1}{2}} du$$

$$= \frac{1}{3} \left[\frac{2}{5} u^{\frac{5}{2}} - 2 \times \frac{2}{3} u^{\frac{3}{2}} \right] + C$$

$$= \frac{2}{15} (x+2)^{\frac{5}{2}} - \frac{4}{9} (x+2)^{\frac{3}{2}} + C \quad (1 \text{ mark})$$

e) (i)

Criteria	Marks
Correctly evaluates the R	1
Correctly evaluates the auxiliary angle α and gives the correct expression $2 \cos (x + \frac{\pi}{6})$	1

$$\begin{aligned}\sqrt{3} \cos x - \sin x &\equiv R \cos (x + \alpha) \\ &= R [\cos x \cos \alpha - \sin x \sin \alpha]\end{aligned}$$

$$\therefore R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = 1$$

$$\therefore R = \sqrt{(\sqrt{3})^2 + 1^2} = 2, \quad R > 0.$$

$$\therefore \cos \alpha = \frac{\sqrt{3}}{2} \quad \text{and} \quad \sin \alpha = \frac{1}{2}$$

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}} \quad \therefore \alpha = \frac{\pi}{6}, \quad 0 < \alpha < \frac{\pi}{2}$$

$$\therefore \sqrt{3} \cos x - \sin x \equiv 2 \cos (x + \frac{\pi}{6})$$

(ii)

Criteria	Marks
Solves the trigonometric expression correctly for $x + \frac{\pi}{6}$	1
Gives $\frac{\pi}{6}$ as the only solution, <i>must</i> mention the domain $0 \leq x \leq \frac{\pi}{2}$.	1

$$\sqrt{3} \cos x - \sin x = 1$$

$$\therefore 2 \cos (x + \frac{\pi}{6}) = 1$$

$$\cos (x + \frac{\pi}{6}) = \frac{1}{2}$$

$$\therefore x + \frac{\pi}{6} = \frac{\pi}{3}, \quad 2\pi - \frac{\pi}{3} = \frac{\pi}{3}, \quad \frac{5\pi}{3} \quad \text{(1 mark)}$$

$$\therefore x = \frac{\pi}{3} - \frac{\pi}{6}, \quad \frac{5\pi}{3} - \frac{\pi}{6}$$

$$= \frac{\pi}{6} \text{ as the only solution as } 0 \leq x \leq \frac{\pi}{2}. \quad \text{(1 mark)}$$

Question 3

a)

Criteria	Marks
• Correctly proves the result for $n = 1$	1
• Correct statement for $n = k$	1
• Adds $(k + 1 + (2(k + 1) - 1))$ to the LHS of the expression for $n = k$ to give the expression for $n = k + 1$	1
• Correct algebraic techniques to produce the RHS as $\frac{1}{2}(k + 1)(3(k + 1) + 1)$.	1
<i>-1 mark without the correct conclusion</i>	

$$\text{Let } S_n = (1 + 1) + (2 + 3) + (3 + 5) + \dots + (n + (2n - 1)) = \frac{1}{2}n(3n + 1)$$

Step 1: Prove S_1 is true.

$$S_1 = (1 + 1) = \frac{1}{2} \times 1(3 \times 1 + 1)$$

$$2 = 2$$

$$\text{LHS} = \text{RHS}$$

$\therefore S_1$ true. (1 mark)

Step2 : Assume S_k is true.

$$S_k = (1 + 1) + (2 + 3) + (3 + 5) + \dots + (k + (2k - 1)) = \frac{1}{2}k(3k + 1)$$

Step2 : Prove S_{k+1} is true.

$$\begin{aligned} \text{LHS } & (1 + 1) + (2 + 3) + (3 + 5) + \dots + (k + (2k - 1)) + (k + 1 + (2(k + 1) - 1)) \quad (1 \text{ mark}) \\ &= \frac{1}{2}k(3k + 1) + (k + 1 + (2(k + 1) - 1)) \\ &= \frac{1}{2}k(3k + 1) + (k + 1) + (2k + 1) \\ &= \frac{1}{2}[k(3k + 1)] + 2(3k + 2) \\ &= \frac{1}{2}[3k^2 + 7k + 4] \\ &= \frac{1}{2}(k + 1)(3k + 4) \\ &= \frac{1}{2}(k + 1)(3(k + 1) + 1) \quad (1 \text{ mark}) \end{aligned}$$

$\therefore S_{k+1}$ is true, given that S_k is true.

Conclusion: We have proved S_1 is true and then using the result from step 3, S_2 is true; hence S_3 is true and so on S_n is true for all integral values of n .

b) (i)

Criteria	Marks
Correctly evaluates the number of arrangements	1

No. of arrangements =

$$\boxed{\text{M}} \quad \underline{3} \quad \underline{2} \quad \underline{1} \quad \boxed{\text{E}} = 6$$

(ii)

Criteria	Marks
Correctly evaluates the number of arrangements	1

Consider OUE as one group. $\boxed{\text{M}} \quad \boxed{\text{OUE}} \quad \boxed{\text{S}}$

The three letters can be organised in 3! Ways and $\boxed{\text{OUE}}$ can be organised in 3! Ways.
 $\therefore$ total number of arrangements = $3! \times 3! = 36$ ways

(iii)

Criteria	Marks
Correctly evaluates the number of arrangements and concludes that the no. of arrangements in part (i) is halved	1
Correctly evaluates the number of arrangements and concludes that the no. of arrangements in part (ii) is halved	1

$$\boxed{\text{M}} \quad \underline{\text{O}} \quad \underline{\text{O}} \quad \underline{\text{S}} \quad \boxed{\text{E}}$$

No. of arrangements = $\frac{3!}{2!} = 3 \therefore$ Answer in part (i) is halved.

$$\underline{\text{M}} \quad \boxed{\text{OOE}} \quad \underline{\text{S}}$$

No. of arrangements = $3! \times \frac{3!}{2!} = 18$. The answer in part (ii) is halved.

c)

Criteria	Marks
• Multiplies both sides of the inequality by $(x - 1)^2$	1
• Correctly manipulates the expression to get $(x - 1)(x^2 + x - 6) \leq 0$	1
• Solves the inequality to get $x \leq -3$; $1 \leq x \leq 2$, $x \neq 1$	1
• Excludes $x = 1$, to get $x \leq -3$; $1 < x \leq 2$	1

$$x + 2 \leq \frac{4}{x - 1}.$$

Multiply by $(x - 1)^2$,

$$(x + 2)(x - 1)^2 \leq 4(x - 1) \quad (\text{1 mark})$$

$$(x + 2)(x - 1)^2 - 4(x - 1) \leq 0$$

$$(x - 1)[(x + 2)(x - 1) - 4] \leq 0$$

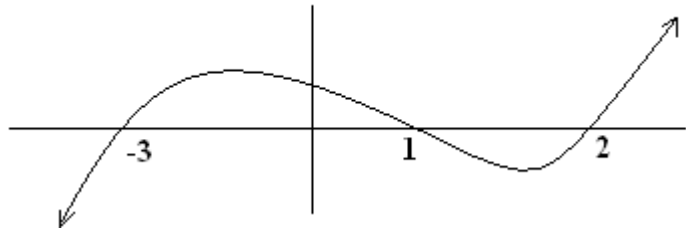
$$(x - 1)(x^2 + x - 2 - 4) \leq 0$$

$$(x - 1)(x^2 + x - 6) \leq 0$$

$$(x - 1)(x + 3)(x - 2) \leq 0$$

$$\therefore x \leq -3; \quad 1 \leq x \leq 2, \quad x \neq 1$$

Solution is $x \leq -3$; $1 < x \leq 2$



Question 4

a)

Criteria	Marks
- Correctly derives the gradient of the normal (<i>must show all steps</i>) - Correctly derives the equation of the normal using the gradient from above.	1 1

$$x = 2at \quad y = at^2$$

$$\therefore \frac{dy}{dx} = \frac{2at}{2a} = t.$$

$\therefore$ Gradient of the normal is $-\frac{1}{t}$. **(1 mark)**

Equation of the normal is

$$y - at^2 = -\frac{1}{t} (x - 2at)$$

$$t(y - at^2) = -x + 2at$$

$$\therefore x + ty = at^3 + 2at. \quad \textbf{(1 mark)}$$

(ii)

Criteria	Marks
- Substitutes $y = 0$ into the equation of the normal to find Q. - Correctly proves the coordinates of R.	1 1

x-intercept: put $x = 0$

$$0 + ty = at^3 + 2at.$$

$$\therefore y = 2a + at^2$$

$$\therefore Q(0, 2a + at^2) \quad \textbf{(1 mark)}$$

$P(2at, at^2)$

$$\therefore \text{The midpoint } R \left(\frac{0 + 2at}{2}, \frac{2a + at^2 + at^2}{2} \right)$$

$$R(at, a + at^2) = R[at, a(1 + t^2)]$$

(iii)

Criteria	Marks
Sets $x = a(1 + t^2)$ and $y = at$ and eliminates t to the Cartesian equation of the locus of R.	1

$$y = a(1 + t^2)$$

$$x = at \Rightarrow t = \frac{x}{a}$$

$$\therefore y = a\left(1 + \left[\frac{x}{a}\right]^2\right)$$

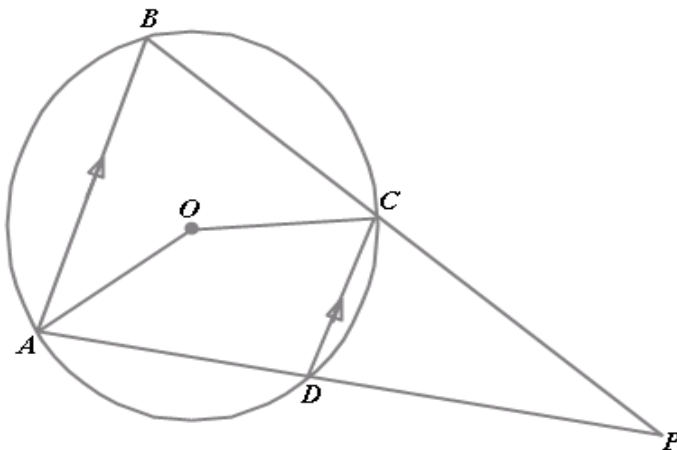
$$= a\left(1 + \frac{x^2}{a^2}\right)$$

$$y = a + \frac{x^2}{a}$$

$$y - a = \frac{y^2}{a} \quad \therefore x^2 = a(y - a) \text{ is a parabola with focal length } \frac{a}{4} \text{ and vertex } (0, a)$$

b)

Criteria	Marks
One mark for each step of working with correct reasoning	3



(i) Let $\angle PDC = \alpha$

$$\therefore \angle PDC = \angle ABC = \alpha$$

(1)

(the exterior angle of a cyclic quadrilateral equals the sum of the opposite interior angles (1 mark))

$$\angle ABC = \angle DCP = \alpha \quad (\text{AB} \parallel \text{DC, corresponding angles in parallel lines are equal})$$

(2)

(1 mark)

ie. $\angle PDC = \angle DCP = \alpha$ (using (1) and (2)) (3)

$\therefore PD = PC$ (using (3) $\triangle PCD$ is isosceles and therefore the sides are equal) (1 mark)

(ii)

Criteria	Marks
One mark for each step of working with correct reasoning	2

$\angle PDC = \alpha$

$\therefore \angle PDC = \angle PAB = \alpha$ ($CD \parallel AB$, corresponding angles in parallel lines are equal) (1 mark)

$\therefore \angle PBA = \angle PAB = \alpha$ (base angles are equal, $\therefore \triangle PAB$ is isosceles) (1 mark)

(iii)

Criteria	Marks
One mark for each step of working with correct reasoning	2

$\angle ABC = \alpha$

$\therefore \angle AOC = 2\alpha$ (angle subtended by arc AC at the centre is double of what is subtended at the circumference.) (1 mark)

In $\triangle PCD$,

$$\begin{aligned}
 \angle CPD &= 180 - (\angle PCD + \angle PDC) \\
 &= 180 - (\alpha + \alpha) \\
 &= 180 - 2\alpha \quad (\text{angle sum of triangle add to } 180^\circ)
 \end{aligned}$$

$\therefore$ In quadrilateral OAPC,

$$\angle CPA + \angle AOC = 180 - 2\alpha + 2\alpha = 180^\circ$$

$\therefore$ OAPC is cyclic. (opposite angles add to 180°) (1 mark)

Question 5

a)

Criteria	Marks
- Correct differentiation of $\tan x$ and x - Applies quotient rule correctly	1 1

$$y = \frac{\tan x}{x}$$

$$u = \tan x \quad u' = \sec^2 x$$

$$v = x \quad v' = 1$$

$$\frac{dy}{dx} = \frac{x \sec^2 x - \tan x}{x^2}$$

b)

Criteria	Marks
- Correct differentiation of $\cos x$ and x - Applies product rule correctly	1 1

$$y = x \cos x$$

$$u = x \quad u' = 1$$

$$v = \cos x \quad v' = -\sin x$$

$$\frac{dy}{dx} = -x \sin x + \cos x$$

c)

Criteria	Marks
Correct integration of $\sin 3x$ and $\cos 2x$	2

$$\int (\sin 3x + \cos 2x) dx = \frac{-\cos 3x}{3} + \frac{\sin 2x}{2} + C$$

d)

Criteria	Marks
- Correct integration - Evaluates the expression correctly	1 1

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sin^2 x \cos x dx &= \left[\frac{\sin^3 x}{3} \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{3} [1 - 0] = \frac{1}{3} \end{aligned}$$

e)

Criteria	Marks
Uses the property of odd functions to evaluate correctly	1

$$\int_{-2\pi}^{2\pi} \cos^2 x \sin^3 x \, dx$$

$$\cos^2 x \sin^3 x \text{ is an odd function; } \therefore \int_{-2\pi}^{2\pi} \cos^2 x \sin^3 x \, dx = 0$$

f)

Criteria	Marks
Integrates $\frac{dy}{dx} = \cos 2x$ correctly	1
Evaluates the constant of integration by substitution	1
Obtains the primitive function and substitutes $x = 2\pi$ to give correct value of y.	1

$$\frac{dy}{dx} = 2 \cos^2 x - 1$$

$$\text{ie. } \frac{dy}{dx} = \cos 2x.$$

$$\therefore y = \frac{\sin 2x}{2} + C \quad (\text{1 mark})$$

$$\text{When } x = \pi \text{ when } y = \pi.$$

$$\therefore \pi = 0 + C \Rightarrow C = \pi \quad (\text{1 mark})$$

$$\therefore y = \frac{\sin 2x}{2} + \pi.$$

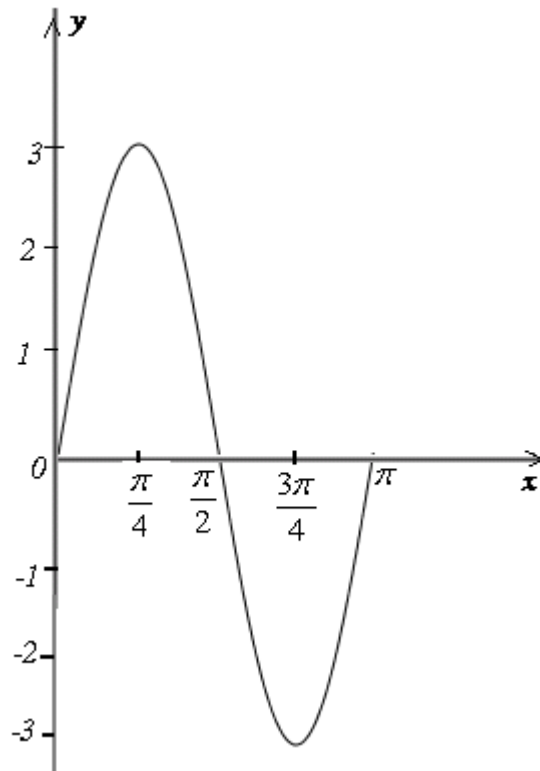
$$\text{When } x = 2\pi, \quad y = 0 + \pi \quad \therefore y = \pi \quad (\text{1 mark})$$

Question 6

a)(i)

Criteria	Marks
Correct amplitude and period	1
Correct shape and uniform scale on the axes and correct labels	1

Amplitude = 3 and period $\frac{2\pi}{2} = \pi$



(ii)

Criteria	Marks
Correct expression to calculate the volume	1
Rewrites $\sin^2 2x$ as $\frac{1 - \cos 4x}{2}$	1
Correct integration and evaluation	1

$$\begin{aligned}
 \text{Volume} &= 2 \times \pi \int_0^{\frac{\pi}{2}} (3 \sin 2x)^2 dx \\
 &= 2 \times \pi \times 9 \times \int_0^{\frac{\pi}{2}} \sin^2 2x dx \quad (1 \text{ mark}) \\
 &= 2 \times \pi \times 9 \times \int_0^{\frac{\pi}{2}} \frac{1 - \cos 4x}{2} dx \quad (1 \text{ mark})
 \end{aligned}$$

$$\begin{aligned}
&= 9\pi \int_0^{\frac{\pi}{2}} 1 - \cos 4x \, dx \\
&= 9\pi \left[x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{2}} \\
&= 9\pi \left[\frac{\pi}{2} - \frac{1}{4} \times 0 - 0 \right] \\
&= \frac{9\pi^2}{2} \text{ unit}^2. \quad \text{(1 mark)}
\end{aligned}$$

b)

Criteria	Marks
Makes x^2 subject of the formula correctly.	1
Gives the expression for volume with y-axis $\pi \int_1^8 16y^{-\frac{2}{3}} dy$	1
Correct integration and evaluation	1

$$x \sqrt[3]{y} = 4 \quad \text{from } y = 1 \text{ to } y = 8$$

$\therefore x$ is positive. The graph is in the 1st quadrant.

$$\begin{aligned}
\text{Volume} &= \int \pi x^2 dy & x \sqrt[3]{y} &= 4 \\
& & x &= \frac{4}{\sqrt[3]{y}} \\
& & \therefore x^2 &= \frac{16}{y^{\frac{2}{3}}} \quad \text{(1 mark)}
\end{aligned}$$

$$\begin{aligned}
\therefore \text{Volume} &= \pi \int_1^8 16y^{-\frac{2}{3}} dy \quad \text{(1 mark)} \\
&= 16\pi \times 3 \times \left[y^{\frac{1}{3}} \right]_1^8 \\
&= 48\pi [2 - 1] \\
&= 48\pi \text{ unit}^3. \quad \text{(1 mark)}
\end{aligned}$$

c)(i)

Criteria	Marks
Differentiates $\tan^3 x$ correctly.	1
Substitutes $\tan^2 x$ as $\sec^2 x - 1$ and completes the proof	1

$$\begin{aligned}
\frac{d}{dx}(\tan^3 x) &= 3 \tan^2 x \cdot \sec^2 x \quad \text{(1 mark)} \\
&= 3(\sec^2 x - 1) \sec^2 x \\
&= 3 \sec^4 x - 3 \sec^2 x \quad \text{(1 mark)}
\end{aligned}$$

(ii)

Criteria	Marks
• Correct solution	2
• Demonstrates the understanding that integration is the opposite operation of differentiation	1

Using the result from part (i),

$$\therefore \int 3 \sec^4 x - 3 \sec^2 x dx = \tan^3 x$$

$$\therefore \int \sec^4 x dx - \int \sec^2 x dx = \frac{\tan^3 x}{3} \quad \text{(1 mark)}$$

$$\int \sec^4 x dx - \tan x = \frac{\tan^3 x}{3}$$

$$\begin{aligned} \int_0^{\frac{\pi}{4}} \sec^4 x dx &= \left[\frac{\tan^3 x}{3} + \tan x \right]_0^{\frac{\pi}{4}} \\ &= \left(\frac{1}{3} + 1 \right) - 0 \\ &= \frac{4}{3} \quad \text{(1 mark)} \end{aligned}$$

Question 7

a) (i)

Criteria	Marks
• One mark for each of the asymptotes	3

$$y = \frac{(2x+1)^2}{4x(1-x)}$$

Vertical asymptotes : $x = 0$ and $x = 1$ (1 mark each)

Horizontal asymptotes : $y = \frac{(2x+1)^2}{4x(1-x)}$

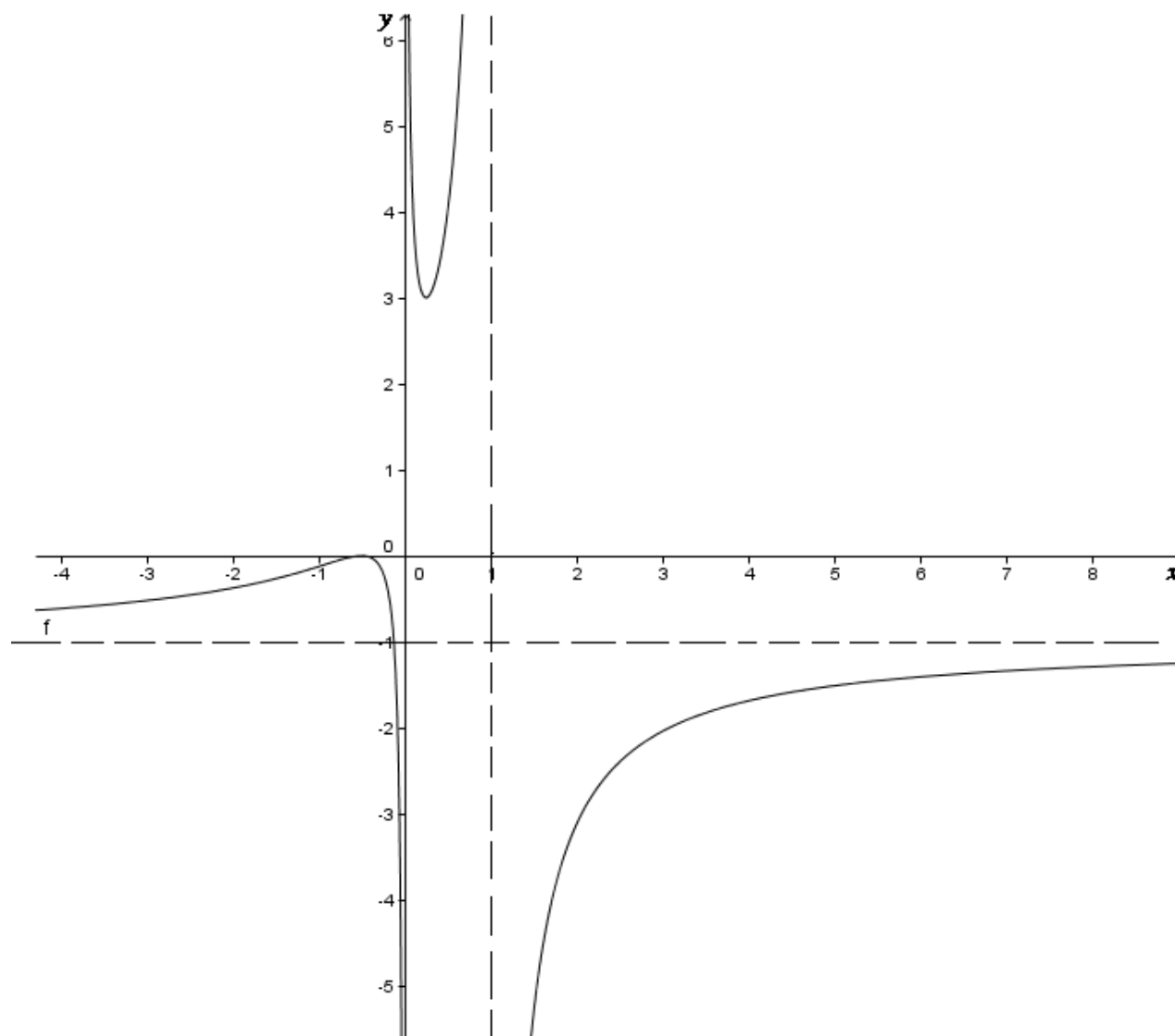
$$y = \frac{(4x^2 + 4x + 1)}{4x - 4x^2}$$

$$\lim_{x \rightarrow \pm\infty} \frac{4 + \frac{4}{x} + \frac{1}{x^2}}{\frac{4}{x} - 4} \rightarrow -1$$

$\therefore y = -1$ is an asymptote. (1 mark)

(ii)

Criteria	Marks
• Appropriate scale and shape	1
• Asymptotes drawn correctly	1
• Turning points labelled correctly	1



b) (i)

Criteria	Marks
• Produces the table of values correctly	1
• Applies Simpson's rule and rounds off correctly	1

$\angle COB = \alpha$. $\therefore \angle OAD = \alpha$. (Corresponding angles in parallel lines are equal $AB \parallel CD$)

$AO = OD = r$ (equal radii)

ΔOAD isosceles (2 equal sides)

$\therefore \angle ADO = \alpha$ (base angles of isosceles triangle are equal)

$\therefore \angle COD = \alpha$. Alternate angles in parallel lines are equal $AB \parallel CD$)

(ii) Area of quadrilateral ABCD = Area of ΔAOD + Area of ΔCOD + Area of ΔBOC

$\angle AOD = 180 - 2\alpha$.

$$\text{Area of } \Delta AOD = \frac{1}{2} r^2 \sin(180 - 2\alpha) = \frac{1}{2} r^2 \sin 2\alpha$$

$$\text{Area of } \Delta COD = \text{Area of } \Delta BOC = \frac{1}{2} r^2 \sin \alpha$$

$$\therefore \text{Area of quadrilateral ABCD, } A = \frac{1}{2} r^2 \sin 2\alpha + 2 \times \frac{1}{2} r^2 \sin \alpha$$

$$A = r^2 \left(\frac{1}{2} \sin 2\alpha + \sin \alpha \right)$$

(iii) Differentiating A with respect to α to find the maximum area,

$$\frac{dA}{d\alpha} = r^2 \left(\frac{1}{2} \times 2 \cos 2\alpha + \cos \alpha \right) = r^2 (\cos 2\alpha + \cos \alpha)$$

$$\text{But } \cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\begin{aligned} \therefore \frac{dA}{d\alpha} &= r^2 (2 \cos^2 \alpha - 1 + \cos \alpha) \\ &= r^2 (2 \cos^2 \alpha + \cos \alpha - 1) \end{aligned}$$

Let $\frac{dA}{d\alpha} = 0$ for possible stationary points.

$$\therefore r^2 (2 \cos^2 \alpha + \cos \alpha - 1) = 0$$

$$(2 \cos^2 \alpha + \cos \alpha - 1) = 0$$

$$(2 \cos \alpha - 1)(\cos \alpha + 1) = 0$$

$$\therefore \cos \alpha = \frac{1}{2}, -1$$

$$\therefore \alpha = \frac{\pi}{3}, \pi, \text{ but } 0 < \alpha < \pi$$

$$\therefore \alpha = \frac{\pi}{3}$$

α	1	$\frac{\pi}{3}$	2
$\frac{dA}{d\alpha}$	$0.12r^2$ /	0 —	$-0.16r^2$ \<

The maximum area of the quadrilateral occurs when $\alpha = \frac{\pi}{3}$ radians.